

MGS-SLAM: Monocular 3D Gaussian splatting SLAM with significance-guided pruning

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Abstract 3D Gaussian splatting demonstrates significant advantages in high-fidelity rendering for *offline* reconstruction of objects or scenes. However, its integration with existing simultaneous localization and mapping (SLAM) systems, especially in monocular scenarios, still suffers from limited localization accuracy and poor rendering quality. In this paper, we propose a new monocular 3D Gaussian splatting SLAM, which achieves high fidelity *online* 3D Gaussian splatting based reconstruction given monocular video input with significance-guided pruning. Our key idea is to maintain structural compactness when optimizing the 3D Gaussians, by adaptively pruning them using a global significance evaluation based on multi-dimensional cues such as visibility, opacity, and volume coefficient. Specifically, we use a frame-to-model pipeline that jointly optimizes camera poses and 3D Gaussians within a sliding-window framework, ensuring a globally consistent 3D Gaussian representation for high-fidelity rendering with significance-guided pruning. Furthermore, an online monocular depth estimation model is incorporated to extract depth priors from input images, to effectively initialize the 3D Gaussian attributes for better camera tracking. Extensive experiments on Replica and TUM datasets demonstrate that our approach substantially improves

both tracking performance and rendering fidelity, and thus provides state-of-the-art results.

Keywords Gaussian splatting; simultaneous localization and mapping (SLAM); pruning

1 Introduction

Visual simultaneous localization and mapping (SLAM) provides fundamental spatial perception capabilities for key applications such as virtual reality (VR), augmented reality (AR), robotics, and unmanned aerial vehicles, demonstrating significant advantages and potential [1–5]. In recent years, SLAM has been increasingly extended to diverse application scenarios, leading to enhanced functionalities such as robust operation in dynamic environments [6, 7], semantic-level mapping [8], and high-quality scene reconstruction [9], which has further strengthened its integration with downstream applications [10]. Among them, online scene reconstruction based SLAM [11–13] aims to simultaneously estimate camera poses and reconstruct three-dimensional (3D) scenes that support free-viewpoint, high-fidelity rendering, thereby enabling highly immersive experiences for applications such as VR and AR [9, 14–17].

Traditional SLAM methods, which represent scenes using primitives such as point clouds [5], meshes [18], or voxels [19], can effectively capture geometric structures but struggle to support high-fidelity rendering [14]. Neural radiance fields (NeRFs) [20], on the other hand, leverage implicit neural networks to build compact and continuous scene representations, enabling free-viewpoint, photorealistic image synthesis, and further enhancing the accuracy of SLAM based scene reconstruction [14, 21]. However, their ray-tracing-based pipeline, which queries volumetric densities

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and colors at numerous points along pixel-derived rays, is extremely time-consuming and cannot support applications that require real-time rendering. Even when using efficient representations such as octrees [22] or hash tables [23], when integrated with SLAM systems, they often suffer from low tracking accuracy or poor reconstruction fidelity [24, 25].

In recent years, 3D Gaussian splatting [26] has significantly improved rendering photorealism and efficiency, and further facilitated downstream tasks like dynamic reconstruction, geometry editing, and physical simulation [27]. It represents scenes using 3D Gaussian ellipsoids, and achieves fast rendering through a differentiable rasterization with adaptive densification. Its integration with SLAM has been extended to various sensing modalities [28], including RGB [29], RGB-D [30], LiDAR [31], and inertial measurement units (IMUs) [32], where the deeply fused sensor data provide richer environmental information, enabling more robust and accurate scene reconstruction. However, monocular SLAM combined with 3D Gaussian splatting still struggles with inaccurate localization and low reconstruction fidelity.

In this paper, our goal is to accurately reconstruct 3D scenes in our 3D Gaussian splatting SLAM framework, MGS-SLAM, from monocular video input, while enabling real-time, high-fidelity rendering using novel significance-guided pruning (see Fig. 1). Unlike previous 3D Gaussian splatting based SLAM systems that maintain accurate 3D Gaussian reconstruction by fusing rich multi-modal cues, our MGS-SLAM uses a significance-guided pruning strategy to retain more significant 3D Gaussians to accurately estimate camera poses, thus enabling high-fidelity online scene reconstruction even from monocular images.

Specifically, we introduce a frame-to-model SLAM framework that jointly optimizes camera poses and 3D Gaussian parameters using a photometric loss, employing a sliding window over keyframes. Moreover, we evaluate the significance of each Gaussian by integrating multi-dimensional cues such as visibility, opacity, and volume coefficients, guiding effective pruning of less significant Gaussians. Furthermore, we employ an online monocular depth estimation model to obtain depth priors from images, which are then combined with color information to initialize the 3D Gaussian attributes including spatial position, opacity, and color.

Using from the above components, our MGS-SLAM achieves accurate scene reconstruction, with both high-fidelity and real-time rendering performance. Furthermore, we have evaluated MGS-SLAM on the widely used Replica [33] and TUM [34] datasets, and compared it to various state-of-the-art monocular scene reconstruction SLAM methods including PhotoSLAM [35], MonoGS [29], and DepthGS [36]. Our experimental results indicate that MGS-SLAM achieves a new state-of-the-art in both tracking and rendering.

2 Related work

2.1 Scene reconstruction

Compared to traditional geometry-based reconstruction methods, NeRFs provide a continuous implicit representation that jointly models geometry and appearance, enabling high-fidelity novel view synthesis with view-dependent effects [20, 37, 38]. Nevertheless, NeRFs' volumetric rendering process is extremely time-consuming, which significantly limits its practical applicability. To further enhance

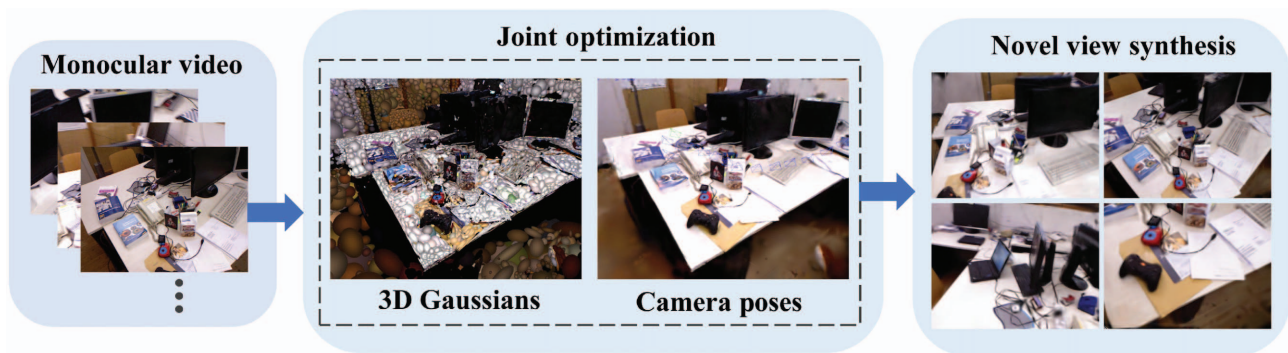


Fig. 1 Given streaming monocular video without camera poses, our MGS-SLAM method jointly optimizes 3D Gaussians and camera poses using a significance-guided pruning strategy, thereby achieving accurate scene reconstruction and high-quality novel view synthesis.

NeRF’s rendering quality, KiloNeRF [39] and BlockNeRF [40] divide the scene into multiple smaller sub-scenes, each represented individually, NISB-MAP [41] extends these by introducing overlapping sub-scenes to improve the consistency and stitching quality in intersecting regions, and SemiNeRF [42] achieves robust camera pose refinement by introducing semantic feature consistency.

In recent years, 3D Gaussian splatting [26] has improved both the rendering fidelity and computational efficiency of NeRFs [20] and their variants [22, 23, 39–41, 43, 44]. Furthermore, some methods have targeted more accurate geometric reconstruction [45–50]. Among them, GESs [46] represents the scene using a set of two-dimensional (2D) opaque surfels with a few 3D Gaussians surrounding them, significantly improving geometric accuracy and enhancing rendering realism; 3DGS-DR [49] uses per-pixel reflection gradients generated by deferred shading to bridge the optimization of neighboring Gaussians, significantly improving non-Lambertian reconstruction quality. In addition, some methods [51–54] have successfully reconstructed vivid human avatars, expanding the application of scene reconstruction in dynamic contexts [55]. Our work was inspired by these works but focuses on monocular online reconstruction SLAM using 3D Gaussian splatting.

2.2 Scene reconstruction SLAM

To support accurate online appearance reconstruction, iMAP [21] first integrates SLAM with NeRFs [20], enabling the joint optimization of camera poses and the implicit map. NICE-SLAM [56] and VoxFusion [57] adopt a voxel-based representation that stores implicit scene information across the entire environment. Furthermore, several studies have integrated implicit representations with more efficient data structures, including octrees [24], hash tables [25], tri-planes [58], and point clouds [59], with certain structures even achieving real-time tracking and reconstruction. Moreover, some NeRF-based SLAM methods have incorporated loop closure [60] and submap strategies [61], enabling more accurate camera localization and globally consistent reconstruction, particularly for long-sequence scenarios. In addition, NeRF-based SLAM methods targeting semantic mapping [62], robustness in dynamic scenes [63], and multi-sensor fusion [64] have further expanded applicability.

Recently, some studies integrating 3D Gaussians with SLAM have demonstrated superior rendering speed and reconstruction accuracy compared to NeRF-based SLAM. For example, Photo-SLAM [35] integrates ORB-SLAM [65] with 3D Gaussian splatting [26], supporting multiple input modalities, including stereo and RGB-D cameras; RTG-SLAM [30] features a compact Gaussian representation and a highly efficient on-the-fly Gaussian optimization scheme, achieving high-quality reconstruction with low memory cost. GPS-SLAM [66] combines colorized signed distance fields with 3D Gaussians, significantly improving reconstruction speed while maintaining high-quality reconstruction, while MonoGS [29] enables relatively robust tracking and reconstruction. In addition, recent studies have also extended 3D Gaussian splatting SLAM to dynamic scenes [67, 68] and semantic segmentation [69]. In contrast, our work newly provides significance-guided pruning for 3D Gaussian splatting based SLAM, which effectively improves the rendering quality especially given monocular input.

3 MGS-SLAM

3.1 Aims and approach

The goal of our MGS-SLAM is to estimate camera poses and reconstruct 3D scenes from monocular input, while achieving real-time, high-fidelity rendering. Figure 2 shows our frame-to-model SLAM framework, which performs joint optimization of camera poses and 3D Gaussians using a sliding-window strategy (see Section 3.2). To maintain structural compactness for the 3D Gaussian optimization, the significance of each Gaussian is quantified by integrating multi-dimensional cues, guiding effective pruning (see Section 3.3). Additionally, keyframe depths are estimated online and subsequently combined with color information to initialize the attributes of the 3D Gaussians (see Section 3.4).

3.2 MGS-SLAM framework

3.2.1 Gaussian splatting

Our MGS-SLAM is represented using 3D Gaussians, each parameterized by opacity α^i and color c_i . The former serves as a weighting factor in volumetric rendering, while the latter is direction-dependent and is modeled using spherical harmonics: color varies

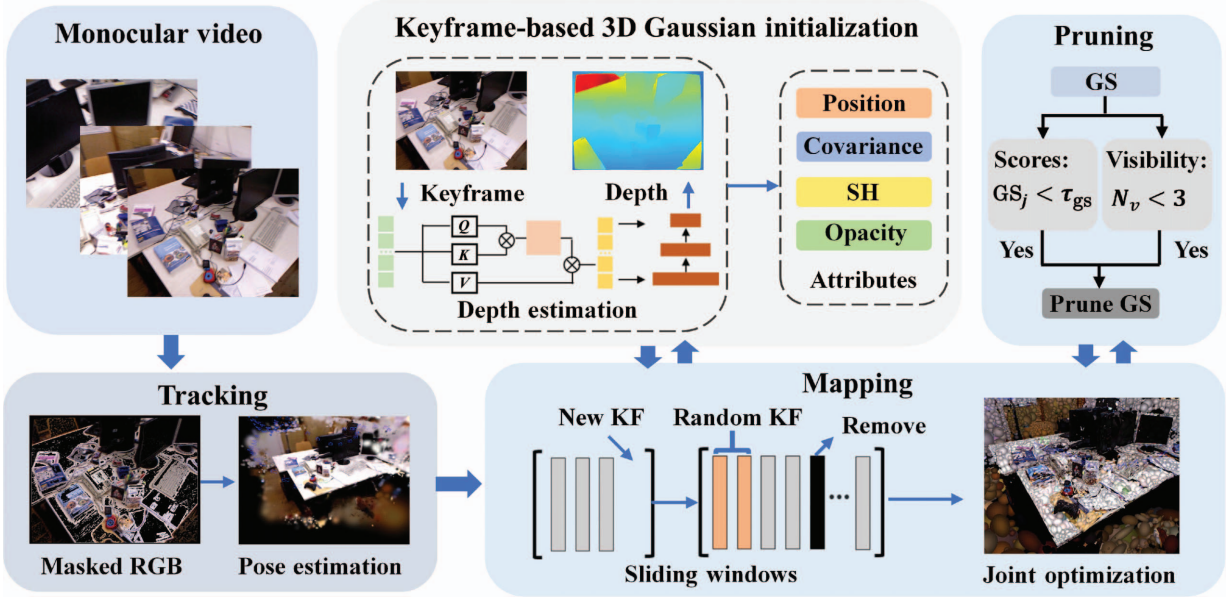


Fig. 2 MGS-SLAM pipeline. Our frame-to-model SLAM framework jointly optimizes camera poses and 3D Gaussian parameters employing a sliding window strategy. We prune less significant Gaussians by evaluating the significance of each Gaussian using multi-dimensional cues. We employ an online monocular depth estimation model to obtain depth priors to efficiently initialize the 3D Gaussian attributes.

according to the viewing direction. In addition, each 3D Gaussian is characterized by a mean μ_W^i and variance Σ_W^i , which in turn define its spatial position and ellipsoidal shape within the scene. The final pixel color c_p is derived through volume rendering, where weighted-color contributions from individual Gaussian ellipsoids are accumulated, as Eq. (1):

$$c_p = \sum_{i \in N} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) \quad (1)$$

where N denotes the number of 3D Gaussians contributing to the rendering, and α_i is obtained by decaying α^i using a 2D Gaussian function. Specifically, the 2D Gaussians $\mathcal{N}(\mu_I, \Sigma_I)$ on the image plane are derived from the 3D Gaussians $\mathcal{N}(\mu_W, \Sigma_W)$ through a projective transformation, as Eq. (2):

$$\mu_I = \pi(T_{CW} \cdot \mu_W), \quad \Sigma_I = J W \Sigma_W W^T J^T \quad (2)$$

where π denotes the projection operator, T_{CW} is the estimated camera pose, J represents the Jacobian matrix that provides a linear approximation of the projection transformation, and W denotes the camera rotation matrix. This computation is fully differentiable, allowing for iterative optimization of the 3D Gaussians' color, scale, orientation, opacity, and position under photometric loss or other relevant constraints.

3.2.2 Tracking

We employ a frame-to-model pipeline for camera pose

optimization, where the photometric loss integrates opacity and image gradient cues:

$$\begin{cases} L_{pt} = E[S(\mathcal{G}, T_{CW}) \odot L_{ptm}] \\ L_{ptm} = \|I(\mathcal{G}, T_{CW})(M_{fd}) - \bar{I}(M_{fd})\|_1 \end{cases} \quad (3)$$

where $S(\mathcal{G}, T_{CW})$ denotes the accumulated opacity of the Gaussians $\mathcal{G}$ from T_{CW} , $I(\mathcal{G}, T_{CW})$ denotes the rendered image, $\bar{I}$ is an observed image, and M_{fd} denotes the Boolean mask obtained from image gradients, which are computed using a Scharr filter. The image is divided into multiple blocks, and M_{fd} is determined for each block according to its median gradient: $M_{fd} = (D_j \geq f D_{mi})$, where D_j denotes the j -th gradient in the i -th block, D_{mi} denotes the mean gradient of the i -th block, and f is a predefined scaling factor.

Pixel regions exhibiting pronounced color variations are identified by M_{fd} , as they generally correspond to robust texture features that contribute to improved tracking accuracy. In addition, the accumulated opacity s_p , computed as

$$s_p = 1 - \prod_{i=1}^N (1 - \alpha_i) \quad (4)$$

is rendered and subsequently used for adjusting the loss weighting. Incorporating the accumulated opacity into pose estimation can mitigate the interference with tracking accuracy caused by inadequate reconstruction. Using the above loss

function, tracking is carried out for N_{iter} iterations, but iterations are stopped early whenever the magnitude of pose update drops below τ_t .

3.2.3 Mapping with significance-guided pruning

In the mapping process, 3D Gaussians are optimized and camera poses are fine-tuned using a sliding-window approach. Specifically, each sliding window comprises the two most recent keyframes along with n additional keyframes whose overlap coefficient (OC) with the current keyframe exceeds a predefined threshold τ_{cko} , where the OC is defined as the intersection of 3D Gaussians observed by the two frames divided by the number of Gaussians observed by the frame with fewer observations. A Gaussian is considered observed by a frame if it is used in the rasterization process and its ray's s_p is below 0.5. To dynamically maintain the window, when the window size exceeds its maximum capacity, the keyframe that exhibits largest global distance from the nearest keyframe while maintaining high spatial redundancy with neighboring views is removed to preserve a more uniform inter-frame spacing within the keyframe distribution. Specifically, we compute the inverse distance between each frame's pose and those of the other frames in the window, summing these distances to quantify spatial redundancy, with higher values indicating greater redundancy. During optimization, to alleviate the issue of catastrophic forgetting, two keyframes randomly sampled from those outside the local window are included in the optimization process. The photometric loss below is used to jointly optimize the 3D Gaussians and p nearest camera poses within the local window.

$$L_{\text{pms}} = E \left[\|I(\mathcal{G}, T_{\text{CW}}) - \bar{I}\|_1 \right] \quad (5)$$

To prevent 3D Gaussians from being abnormally elongated in certain directions due to insufficient viewing constraints, an anisotropy penalty F_{iso} is imposed on their scales,

$$F_{\text{iso}} = E \left[\sum_{i=1}^{|\mathcal{G}|} \|s_{3d} - \bar{s}_{3d}\|_1 \right] \quad (6)$$

where s_{3d} denotes the 3 scales of each 3D Gaussian, $\bar{s}_{3d}$ represents the mean value of the 3 scales. This serves to promote more geometrically consistent scene representations and improve the robustness of 3D Gaussian rendering during novel-view synthesis. Accordingly, the final loss function is given by

$$L_{\text{pm}} = \sum_{\forall k \in \mathcal{W}} L_{\text{pms}}^k + \lambda_{\text{iso}} F_{\text{iso}} \quad (7)$$

where $\mathcal{W}$ represents the set of keyframes within the local window together with two randomly selected keyframes, k denotes the index of a keyframe within $\mathcal{W}$, and λ_{iso} is a weighting factor.

For 3D Gaussian splatting SLAM, the incorporation of keyframe-based 3D Gaussian initialization facilitates efficient scene reconstruction and enhanced representation of fine-grained details. However, although increasing the number of 3D Gaussian primitives enables the preservation of richer scene information, it simultaneously leads to a substantial increase in number of parameters during mapping and optimization, thereby imposing heavy memory demands and compromising the computational efficiency of the SLAM system. Moreover, unavoidable popping artifacts degrade the fidelity of the reconstructed 3D scene and can substantially impair the tracking performance of frame-to-model methods. Traditional pruning strategies, which evaluate 3D Gaussians based solely on opacity, are insufficient for comprehensively measuring their significance. Consequently, locally detailed or structurally complex regions may lose critical Gaussian primitives, which can adversely affect both tracking stability and overall reconstruction quality. To address this, we propose the use of a global significance score to quantify the significance of 3D Gaussian primitives and develop a corresponding pruning strategy to integrate it into the backend of our visual SLAM framework, following adaptive densification.

3.3 Significance-guided pruning

3.3.1 Global significance scores

In the backend, global significance scores are computed for all keyframes within the local window. Specifically, for each keyframe, the 3D Gaussians within the view frustum are projected onto the pixel plane via a rasterizer, and their contributions to pixel rendering are computed. By incorporating the opacity and scale attributes of each Gaussian, the global significance scores are then derived, as Eq. (8):

$$\text{GS}_j = \sum_{i=1}^{mhw} 1(\mathcal{G}_j, r_i) \cdot \alpha_j \cdot \prod_{k=1}^{j-1} (1 - \alpha_k) \cdot \gamma(\Sigma_j) \quad (8)$$

where m denotes the number of camera frames used for parameter computation, h and w denote the image height and width respectively, $\mathcal{G}_j$ denotes the j -th Gaussian in the 3D scene, and $1(\mathcal{G}_j, r_i)$

indicates whether the j -th Gaussian contributes to the rendering of the i -th pixel, with a value of 1 if it contributes and 0 otherwise. $\alpha_j \cdot \prod_{k=1}^{j-1} (1 - \alpha_k)$ is the weight of the j -th Gaussian contributing to the rendering of the i -th pixel. Since a single Gaussian may participate in the rendering of multiple pixels, all pixels are traversed to accumulate the number of times each primitive contributes. In the computation of global significance scores, a 3D Gaussian is considered more significant if it contributes greater weight across a larger number of rendered pixels. Moreover, since a higher-volume 3D Gaussian generally contributes more to rendering for both the training viewpoints and novel views, it is therefore typically more significant. However, using Gaussian volume alone tends to overemphasize background Gaussians, leading to excessive pruning of Gaussians modeling fine geometry. To address this issue, $\gamma(\Sigma_j)$ is computed based on the scale of each 3D Gaussian, thereby modulating the influence of its volume on the global significance score robustly:

$$\gamma(\Sigma_j) = (V_{\text{norm}})^\beta, V_{\text{norm}} = \min\left(\frac{V(\Sigma)}{V_{\text{max } 90}}, 1\right) \quad (9)$$

where $V_{\text{norm}} \in (0, 1]$ denotes the normalized volume of the 3D Gaussian, β is a hyperparameter, $V_{\text{max } 90}$ denotes the volume at the 90th percentile of all 3D Gaussians in the scene, after sorting them in ascending order by volume. Furthermore, $V(\Sigma) = 4\pi abc/3$, where a , b , and c denote the semi-axes of the 3D Gaussian.

3.3.2 Preventing catastrophic map forgetting

Global significance scores provide a means to assess the significance of 3D Gaussian primitives; however, using these scores as the sole criterion for pruning can lead to substantial forgetting, since the local window's field of view may not fully capture the global scene. Although this could be mitigated by rendering from all viewpoints rather than only those in the local window, to obtain a more globally representative significance measure, such an approach would incur considerable computational overhead. To address this, we directly preserve 3D Gaussians with opacity above a threshold τ_α as they typically encode key structural elements of the scene, as well as primitives with participation counts below a threshold τ_c , which may fall outside the local window's view frustum. Specifically, after excluding the primitives exempted from significance-guided

pruning, the remaining Gaussians are sorted by their global significance scores, and pruning is applied using a pre-defined ratio τ_r . This approach enables a more efficient integration of significance-guided pruning into our visual SLAM pipeline, effectively mitigating map forgetting while maintaining a compact yet faithful 3D Gaussian splatting scene.

3.3.3 Co-visibility-guided pruning

For monocular 3D Gaussian splatting SLAM, the optimized 3D Gaussian primitives often exhibit noticeable rendering artifacts due to the lack of geometric priors, which can severely compromise both tracking and reconstruction fidelity. This issue is especially pronounced for Gaussians inserted by newly selected keyframes, as they are typically constrained by fewer viewpoint observations. To address this, we perform co-visibility-guided pruning within the local window after backend optimization in accordance with multi-view consistency. Specifically, visibility checks are conducted across all keyframes in the local window, and any 3D Gaussian observed by at most two keyframes and also inserted by the three most recent keyframes is pruned. This strategy allows newly inserted Gaussians to more robustly encode the scene while mitigating the impact of multi-view inconsistencies on tracking performance.

3.4 Keyframe-based 3D Gaussian initialization

3.4.1 New keyframe decision

After obtaining an initial estimate of the camera pose in the frontend, a keyframe selection strategy is employed to determine whether to add new keyframes. Once selected, the new keyframe is incorporated into the current local window, where joint optimization of the 3D Gaussians and camera poses is conducted in the backend. In this process, overly sparse keyframes may lead to insufficient viewpoint coverage, hindering the accurate optimization of 3D Gaussians, whereas overly dense keyframes increase computational overhead and may cause greater accumulation of tracking errors. Keyframe selection is performed by jointly considering frame interval, co-visibility ratio, and inter-frame distance, where co-visibility ratio is defined as the intersection of 3D Gaussians observed by the two frames divided by their union. Specifically, a frame is selected as a keyframe if it satisfies either of the following conditions: (i) the translational distance to its nearest keyframe exceeds $\tau_{\text{tmax}} \hat{D}_m$, where $\hat{D}_m$



is the median value of the depth map rendered for the current frame, and the frame is separated by at least τ_f frames, or (ii) the translational distance exceeds $\tau_{tmin} \hat{D}_m$, the co-visibility ratio exceeds τ_{ck} , and the frame interval is at least τ_f frames. By jointly considering inter-frame distance and co-visibility ratio, the proposed criterion captures both camera motion at the pose estimation level and variations in scene coverage at the reconstruction level, facilitating more accurate and efficient 3D scene reconstruction.

3.4.2 3D Gaussian initialization

Monocular visual SLAM systems, which rely solely on color information for pose estimation, are prone to tracking drift due to inconsistencies across multiple views. When combined with 3D Gaussian splatting, the lack of geometric priors further limits reconstruction accuracy, as the system has insufficient constraints to reliably recover the 3D scene structure. To address these issues, we propose a depth-enhanced monocular visual SLAM framework, which tightly integrates a monocular depth estimation network into the system frontend. This integration compensates for the inherent lack of depth observations and improves both tracking and mapping performance. Specifically, we employ the DPT [70] monocular depth estimation model pre-trained on the Omnidata scanning dataset, which provides high computational efficiency while producing robust and accurate depth predictions for diverse scenes. To maintain efficiency, depth estimation is performed only on selected keyframes. Using the keyframe images and their corresponding depth maps, along with the preliminarily estimated camera poses and intrinsic parameters, the 3D spatial coordinates and color attributes of pixels are computed via back-projection. Random downsampling is then applied to selected points for the center positions and color attributes of 3D Gaussian primitives, while the scale parameters of each Gaussian are initialized based on distances to their nearest neighbors using a KNN-based approach. By providing accurate depth priors for the initialization of 3D Gaussians, the proposed framework improves both the efficiency and the quality of scene reconstruction.

4 Evaluation

We have conducted a comprehensive evaluation of our MGS-SLAM, using both qualitative evaluation

and quantitative comparisons, with real-world and synthetic datasets, for 3D Gaussian splatting-based reconstruction and camera localization. Ablation studies were also conducted to highlight the contributions of the depth estimation module and the significance-guided Gaussian pruning strategy.

4.1 Experimental setup

4.1.1 Datasets

We evaluated MGS-SLAM on the widely adopted real-world TUM [34] dataset, which was captured using a Microsoft Kinect sensor and contains diverse indoor scenes with rich geometric structures, photometric variations, and moderately dynamic objects. Furthermore, we conducted experiments on the synthetic Replica [33] dataset, which offers high-resolution textures and extensive camera motion trajectories, enabling a more thorough evaluation.

4.1.2 Implementation details

Our experiments were conducted on a computer equipped with a 2.4 GHz Intel Core i9-12900F CPU and a single NVIDIA GeForce RTX 4090 GPU. For tracking, the number of iterations N_{iter} was set to 100 on the TUM [34] dataset and 3000 on the Replica [33] dataset, with a convergence threshold of $\tau_t = 10^{-4}$ in both cases. Mapping optimization was performed for 150 iterations on TUM [34] and 300 iterations on Replica [33]. Local window maintenance was configured to preserve spatial coverage while controlling redundancy: for TUM, we set $\tau_{cko} = 0.3$, $n = 6$, and $p = 3$; for Replica, $\tau_{cko} = 0.4$, $n = 8$, and $p = 5$. Keyframe selection was governed by frame interval, co-visibility ratio, and inter-frame distance: for TUM, we set $\tau_f = 5$, $\tau_{ck} = 0.9$, $\tau_{tmax} = 0.08$, and $\tau_{tmin} = 0.05$; for Replica, $\tau_f = 2$, $\tau_{ck} = 0.975$, $\tau_{tmax} = 0.02$, and $\tau_{tmin} = 0.01$. Pruning of 3D Gaussians was controlled by $\tau_\alpha = 0.7$, $\tau_c = 5$, $\beta = 0.1$, with pruning ratio $\tau_r = 0.85$ for TUM and $\tau_r = 0.95$ for Replica. The mapping loss function incorporated an anisotropy penalty weight $\lambda_{iso} = 10$ for both datasets. These settings ensured stable convergence and high-fidelity reconstruction in our pipeline.

4.1.3 Metrics

In our experiments, camera pose estimation was quantitatively evaluated using root mean square error (RMSE) of the absolute trajectory error (ATE) computed over all selected keyframes. To faithfully assess the fidelity of the reconstructed



3D scene, we employed standard photometric metrics, including peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and learned perceptual image patch similarity (LPIPS), following the evaluation protocol established by MonoGS [29].

4.1.4 Baseline methods

To comprehensively assess the performance of our method, we used the two benchmark datasets to conduct quantitative and qualitative comparisons to MonoGS [29] and DepthGS [36] using their publicly available implementations, as they represent state-of-the-art methods for monocular 3D Gaussian splatting SLAM. Furthermore, we included comparisons to other advanced methods, such as NICER-SLAM [73], Photo-SLAM [35], and Q-SLAM [75] for a broader evaluation.

4.2 Quantitative evaluation

4.2.1 Camera tracking accuracy

Table 1 reports the tracking results of our method on the TUM [34] dataset, together with a comparison to state-of-the-art methods. Our method achieves

Table 1 Camera tracking results using the TUM dataset. ATE RMSE is reported in cm

Method	fr1/d	fr2/x	fr3/o	Avg.
DSO [71]	22.40	1.10	9.50	11.00
ORB-SLAM3 [3]	4.33	10.46	123.23	46.01
DepthCov-VO [4]	5.60	1.20	68.80	25.20
DROID-VO [5]	5.20	10.70	7.30	7.73
MonoGS [29]	3.80	4.65	3.60	4.02
DepthGS [36]	9.67	—	—	—
Ours	2.09	3.35	2.10	2.51

accurate tracking across the three sequences, yielding the best average performance and attaining the highest accuracy on fr1/desk and fr3/office. Although the performance on fr2/xyz is slightly lower than that of DSO [71], our system surpasses MonoGS [29], which is also a Gaussian splatting based SLAM framework employing a frame-to-model optimization pipeline. Moreover, compared to DepthGS [36], a frame-to-frame SLAM system built upon the 3D Gaussian splatting representation, our approach achieves significantly higher tracking accuracy on fr1/desk, demonstrating the advantage of our pose estimation strategies.

Table 2 reports tracking results for our method on the Replica [33] dataset, together with a comparison to state-of-the-art methods. Our method achieves the highest accuracy on six sequences and the best average performance. The lower performance on the o4 and r1 sequences may be attributed to the challenging camera trajectories, which present difficulties for joint optimization of camera pose and scene representation, yet the method still outperforms MonoGS [29].

4.2.2 Rendering performance

Tables 3 and 4 report the rendering performance of our method on the TUM [34] dataset, together with a comparison to state-of-the-art methods. Our MGS-SLAM achieves the highest PSNR and SSIM scores across all three sequences and is comparable to Photo-SLAM [35] for LPIPS, demonstrating its superior rendering fidelity compared to existing

Table 2 Camera tracking results using the Replica dataset. ATE RMSE is reported in cm

Method	o0	o1	o2	o3	o4	r0	r1	r2	Avg.
ORB-SLAM3 [3]	110.21	103.95	65.36	51.15	1.19	51.39	26.38	4.33	51.75
DROID-SLAM [5]	53.27	34.43	119.31	98.09	83.73	103.89	53.15	66.94	76.60
MonoGS [29]	56.54	24.93	41.59	12.29	81.78	10.15	68.27	29.74	40.66
DepthGS [36]	22.00	11.00	—	—	—	—	10.50	—	—
Ours	3.12	1.99	1.37	2.03	11.41	1.21	35.65	1.05	7.23

Table 3 Average rendering performance of our method compared to state-of-the-art methods on the TUM and Replica datasets

Dataset	Metric	GO-SLAM [72]	NICER-SLAM [73]	MoD-SLAM [74]	Photo-SLAM [35]	MonoGS [29]	Q-SLAM [75]	DepthGS [36]	Ours
TUM	PSNR↑	—	—	—	19.68	21.88	—	17.16	22.36
	SSIM↑	—	—	—	0.693	0.731	—	0.654	0.745
	LPIPS↓	—	—	—	0.268	0.330	—	0.487	0.317
Replica	PSNR↑	22.13	25.41	27.31	29.28	29.25	32.49	19.92	34.53
	SSIM↑	0.730	0.827	0.850	0.883	0.890	0.890	0.678	0.932
	LPIPS↓	—	0.191	—	0.139	0.210	0.170	0.478	0.134

Table 4 Rendering performance of our method compared to state-of-the-art methods using the TUM dataset

Method	Metric	fr1/d	fr2/x	fr3/o
Photo-SLAM [35]	PSNR↑	18.81	21.35	18.88
	SSIM↑	0.681	0.727	0.672
	LPIPS↓	0.329	0.187	0.289
Mono GS [29]	PSNR↑	21.10	22.43	22.11
	SSIM↑	0.707	0.727	0.760
	LPIPS↓	0.351	0.286	0.353
Depth GS [36]	PSNR↑	17.72	18.30	15.45
	SSIM↑	0.707	0.741	0.515
	LPIPS↓	0.436	0.391	0.635
Ours	PSNR↑	21.39	23.22	22.47
	SSIM↑	0.715	0.750	0.770
	LPIPS↓	0.351	0.258	0.343

methods. The relatively superior LPIPS performance of Photo-SLAM [35] can be attributed to its multi-scale pyramid representation. It is noteworthy that our method is fully compatible with hierarchical representations and could be extended with a hierarchical 3D Gaussian splatting strategy [76, 77] to further enhance the expressive capacity of the scene model. This direction will be explored in future work.

The rendering performance of our method on the Replica [33] dataset is further compared in Tables 3 and 5 to other state-of-the-art methods. Our approach achieves nearly the best performance across all sequences on the three rendering metrics and attains the highest average scores, demonstrating its clear advantage in capturing scene details and producing high-fidelity rendering.

4.2.3 Timing analysis

To quantify the computational performance of our

system, we measured the total runtime required to process all frames of the TUM [34] fr2/xyz dataset, and computed the average processing time by dividing the total runtime by the number of frames. Furthermore, all scene images were rendered sequentially, and the average rendering speed was computed in the same manner. As Table 6 reports, our system supports tracking and reconstruction at 3.04 frames/s, while achieving a rendering speed of 926.98 frames/s. This demonstrates that our method effectively utilizes the rendering efficiency of 3D Gaussian splatting and enables scene reconstruction at a considerable speed.

4.3 Qualitative results

4.3.1 Camera tracking trajectories

Figures 3 and 4 visualize some of our camera tracking trajectories and compare them to those from MonoGS [29], which also uses a frame-to-model pipeline and achieves high tracking accuracy. On the TUM dataset [34], our method follows the ground-truth camera trajectory (shown as dashed lines) more closely, demonstrating accurate tracking. Furthermore, on the Replica [33] dataset, we provide a more extensive comparison of camera trajectories. Our method robustly achieves accurate tracking across trajectories with various types of motion.

4.3.2 Qualitative rendering comparison

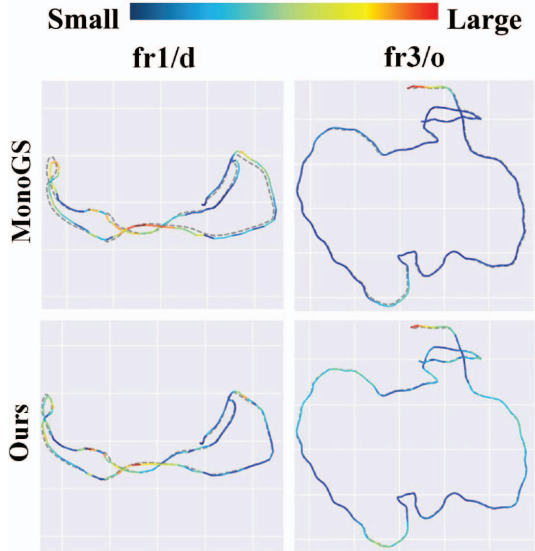
Figures 5 and 6 compare rendering results across all sequences using public implementations of two state-of-the-art methods, MonoGS [29] and DepthGS [36]. Our method robustly reconstructs all scenes with high

Table 5 Rendering performance of our method compared to state-of-the-art methods on the Replica dataset

Method	Metric	o0	o1	o2	o3	o4	r0	r1	r2
NICER-SLAM [73]	PSNR↑	28.54	25.86	21.95	26.13	25.47	25.33	23.92	26.12
	SSIM↑	0.866	0.852	0.820	0.856	0.865	0.751	0.771	0.831
	LPIPS↓	0.172	0.178	0.195	0.162	0.177	0.250	0.215	0.176
Photo-SLAM [35]	PSNR↑	34.42	32.38	28.08	28.06	30.40	26.42	27.08	27.43
	SSIM↑	0.940	0.904	0.900	0.886	0.921	0.787	0.841	0.889
	LPIPS↓	—	0.113	0.131	0.132	0.109	0.221	0.177	0.138
MonoGS [29]	PSNR↑	34.59	34.59	26.59	29.28	27.13	27.95	26.29	27.55
	SSIM↑	0.932	0.933	0.888	0.906	0.902	0.857	0.813	0.888
	LPIPS↓	0.167	0.170	0.240	0.165	0.251	0.168	0.268	0.247
DepthGS [36]	PSNR↑	24.87	26.69	19.23	13.48	14.20	21.54	22.29	17.03
	SSIM↑	0.796	0.869	0.772	0.421	0.521	0.784	0.753	0.508
	LPIPS↓	0.399	0.306	0.377	0.681	0.612	0.391	0.420	0.638
Ours	PSNR↑	36.71	39.00	35.24	34.39	32.14	32.75	29.02	36.96
	SSIM↑	0.949	0.957	0.941	0.939	0.937	0.928	0.851	0.956
	LPIPS↓	0.131	0.122	0.129	0.111	0.160	0.104	0.215	0.099

Table 6 Timing analysis

System		Rendering	
Time (s)	Frame (s^{-1})	Time (s)	Frame (s^{-1})
1119.13	3.04	3.67	926.98

**Fig. 3** Camera trajectory comparison on the TUM dataset.

fidelity, capturing richer and more realistic details than these baselines.

4.4 Ablation analysis

To evaluate the effectiveness of our strategies, we conducted ablation experiments on the TUM [34] and Replica [33] datasets. Specifically, for the variant without significance-guided pruning (w/o SP), 3D Gaussians with opacity lower than 0.7 were pruned.

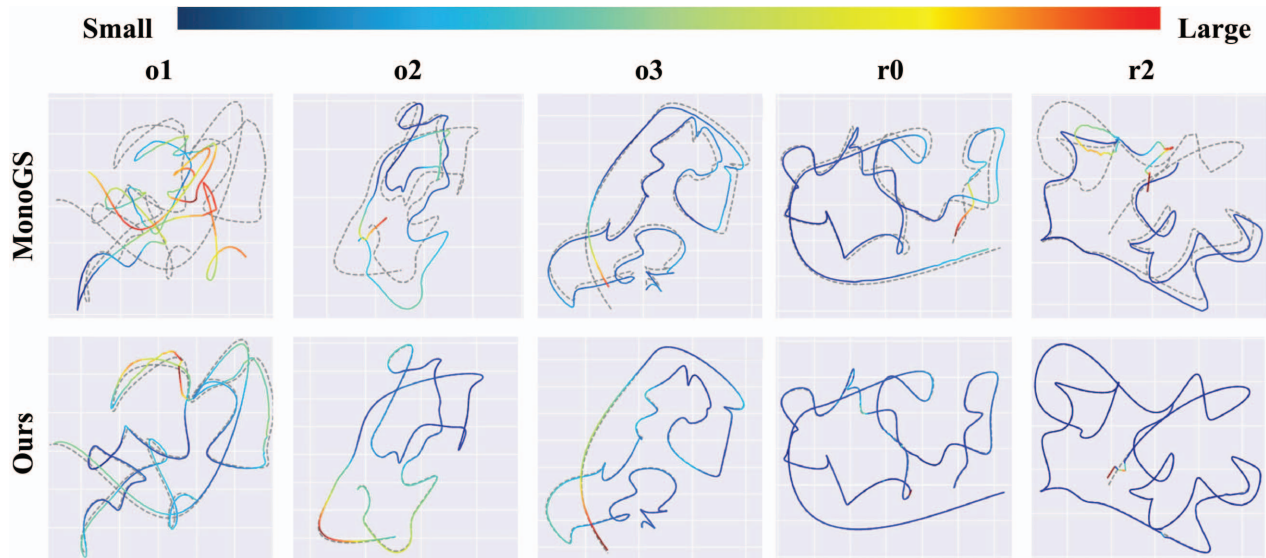
To ablate the 3D Gaussian initialization (w/o DE), the depth estimation network was not used; instead, depth values were directly set to 2 with added random noise in the range of ± 0.3 . To ablate co-visibility-guided pruning (w/o MP), we preserved all newly inserted Gaussians. In addition to the DPT [70] depth estimation network primarily evaluated in this work, we also conducted experiments using DA3 [78] to validate the effectiveness of incorporating a depth estimation network for Gaussian initialization. As Table 7 shows, all of our strategies significantly improved tracking accuracy; Table 8 further shows the improvements in rendering quality. Moreover, we provide a qualitative evaluation of the rendering provided by these variants on the fr2/x sequence of the TUM [34] dataset in Fig. 7, further demonstrating the effectiveness of our strategies.

4.5 Limitations

While our MGS-SLAM demonstrates strong performance in both camera tracking and 3D scene reconstruction, it still exhibits several

Table 7 Tracking results in ablation study using TUM and Replica datasets. ATE RMSE is reported in cm

Method	fr1/d	fr2/x	fr3/o	r0
w/o SP & w/o DE	3.66	4.61	3.58	10.07
w/o SP	4.62	3.27	2.70	2.96
w/o DE	3.09	4.42	3.04	14.11
w/o MP	82.42	3.54	6.45	2.21
Ours (DA3)	2.20	0.88	2.70	1.22
Ours (DPT)	2.09	3.35	2.10	1.21

**Fig. 4** Camera trajectory comparison on the Replica dataset.

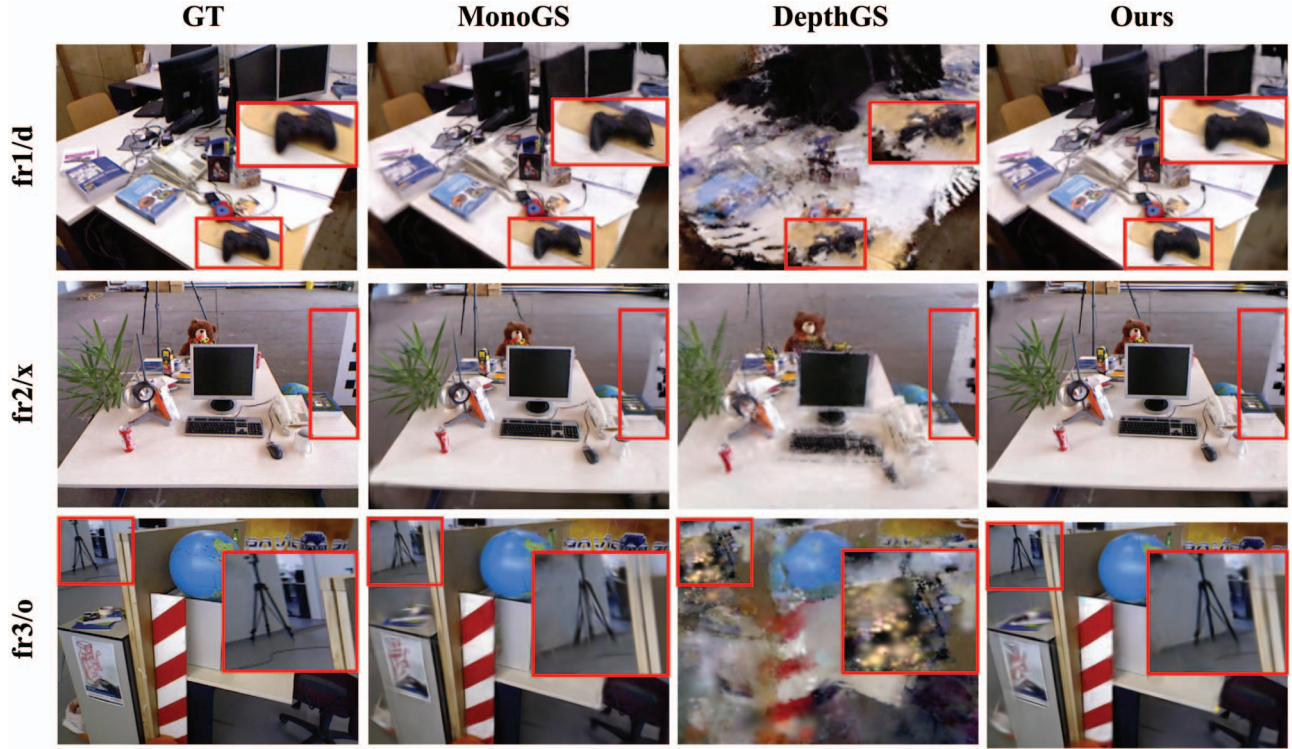


Fig. 5 Comparison of rendering results for our MGS-SLAM and other methods using the TUM dataset.

Table 8 Rendering performance in ablation study using TUM and Replica datasets

Method	Metric	fr1/d	fr2/x	fr3/o	r0
w/o SP & w/o DE	PSNR↑	21.10	22.54	22.30	29.69
	SSIM↑	0.707	0.730	0.766	0.897
	LPIPS↓	0.351	0.284	0.348	0.155
w/o SP	PSNR↑	21.19	22.85	22.42	30.06
	SSIM↑	0.713	0.746	0.767	0.886
	LPIPS↓	0.348	0.266	0.345	0.153
w/o DE	PSNR↑	21.15	22.37	22.39	30.09
	SSIM↑	0.709	0.731	0.768	0.905
	LPIPS↓	0.353	0.282	0.346	0.148
w/o MP	PSNR↑	17.22	23.17	21.82	31.97
	SSIM↑	0.600	0.745	0.756	0.920
	LPIPS↓	0.487	0.259	0.362	0.109
Ours (DA3)	PSNR↑	21.39	24.73	22.49	32.52
	SSIM↑	0.717	0.797	0.769	0.924
	LPIPS↓	0.334	0.212	0.340	0.102
Ours (DPT)	PSNR↑	21.39	23.22	22.47	32.75
	SSIM↑	0.715	0.750	0.770	0.928
	LPIPS↓	0.351	0.258	0.343	0.104

limitations. The current frame-to-model incremental reconstruction strategy ensures higher geometric consistency, but incurs increased computational cost for pose estimation compared to frame-to-frame methods. Importantly, our 3D Gaussian initialization and significance-guided pruning strategies are fully

compatible with frame-to-frame tracking frameworks, offering potential for future integration to balance efficiency and fidelity. As Fig. 8 shows, rendering results on the ScanNet++ [79] sequence indicate that our MGS-SLAM may be challenged by scenes containing prominent non-Lambertian surfaces, owing to the degradation of multi-view consistency and the inherent limitations of 3D Gaussians in modeling color anisotropy. Incorporating semantic priors or reflectance-aware modeling may improve robustness in such non-ideal scenes and enhance the overall fidelity of the reconstructed 3D Gaussians. We leave these directions for future exploration.

5 Conclusions

We present MGS-SLAM, a system that leverages a frame-to-model pipeline that jointly optimizes camera poses and 3D Gaussians within a sliding-window framework, ensuring globally consistent 3D Gaussian representations for high-fidelity rendering with the significance-guided pruning. Experimental results on the public datasets demonstrate that our method significantly improves both tracking accuracy and rendering fidelity, outperforming existing monocular SLAM methods. We hope this work will inspire



Fig. 6 Comparison of rendering results for our MGS-SLAM and other methods using the Replica dataset.



Fig. 7 Ablation study rendering results on the fr2/x sequence from the TUM dataset.



Fig. 8 Rendering results using the ScanNet++ dataset, with prominent non-Lambertian regions highlighted.

further research on more robust and accurate monocular SLAM systems, broadening their potential applications in real-time 3D perception.

Availability of data and materials

TUM, Replica, and ScanNet++ are publicly released datasets.

Author contributions

Tongtai Cao and Kaiyue Guo proposed and executed the key idea, conducted the main experiments, and contributed to the paper writing. Hao Sha participated in the method design and helped refine the writing. Deqi Li explored the method design and experimental setup. Shi-Sheng Huang provided guidance and feedback on the main idea, experimental results, and helped improve the final manuscript. Yue Liu led the project, contributed to the method design and overall paper writing.

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Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article.

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